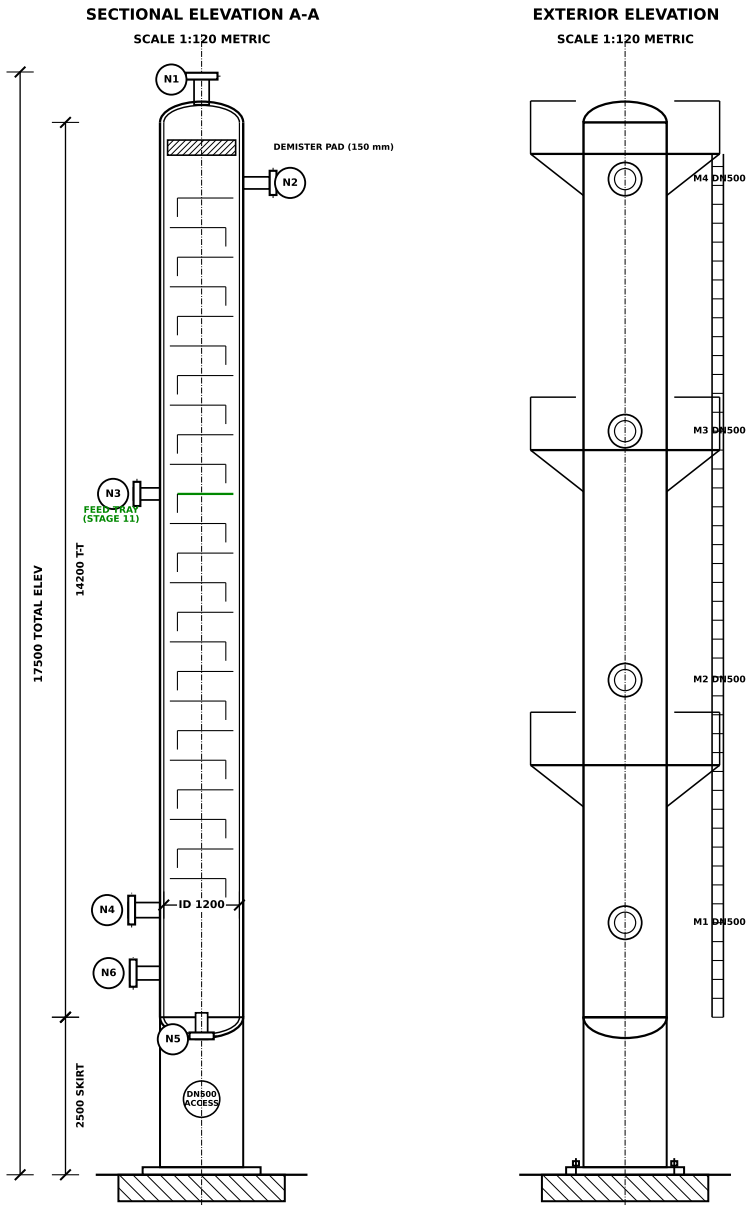

TABLE 1: NOZZLE & MANWAY SCHEDULE (ASME B16.5 CLASS 150 RF)

NOZ	DN (mm)	RATING	FACING	SERVICE / DESTINATION
N1	250 (10")	CL 150	RF	Overhead Vapor to Condenser (E-102)
N2	100 (4")	CL 150	RF	Reflux Return to Tray 1
N3	80 (3")	CL 150	RF	Preheated Feed to Tray 11 (Bubble Pt)
N4	300 (12")	CL 150	RF	Reboiler Vapor Return from E-103
N5	80 (3")	CL 150	RF	Bottoms Liquid Draw to Heat Exchanger
N6	300 (12")	CL 150	RF	Reboiler Liquid Feed to E-103
M1-M4	500 (20")	CL 150	RF	Manways (Sump, Feed, Mid, Top)
PSV-1	100 (4")	CL 150	RF	Overpressure Safety Relief to Flare
VENT	50 (2")	CL 150	RF	Nitrogen Purge / Startup Vent
DRAIN	50 (2")	CL 150	RF	Column Bottom Drain / Cleanout

TABLE 2: PRESSURE VESSEL SPECIFICATIONS (ASME SEC VIII DIV 1)

PARAMETER	VALUE	UNIT	STANDARD / REMARKS
Design Pressure	0.35 / FV	MPa(g)	Full vacuum rated for steam-out
Design Temperature	130.0	C	Maximum boiling conditions
Operating Pressure (Top)	101.325	kPa(a)	Atmospheric rectification
Operating Temperature (Top)	78.15	C	Ethanol-water vapor temperature
Operating Temperature (Bottom)	99.8	C	Boiling water bottoms
Vessel Inside Diameter	1200	mm	Uniform ID throughout column
Tangent-to-Tangent Height	14,200	mm	Active tray section + disengagement
Overall Vessel Height	17,500	mm	Including skirt and ellipsoidal heads
Shell / Head Material	SA-240 304L	-	Corrosion resistant austenitic SS
Head Geometry	2:1 Ellipsoidal	-	ASME BPVC Section VIII Division 1
Corrosion Allowance	1.5	mm	Organic chemical service
Skirt Support Material	SA-516 Gr 70	-	Structural carbon steel skirt
Radiography Examination	100 % (Full)	-	Joint efficiency E = 1.0
Hydrostatic Test Pressure	0.525	MPa(g)	1.5 x Design Pressure

JADAVPUR UNIVERSITY | CHEMICAL ENGINEERING DEPARTMENT | PROCESS EQUIPMENT DESIGN & DRAWING

DRAWING TITLE	LAB / SUBJECT	COURSE / BRANCH	SCALE
DISTILLATION COLUMN GENERAL ARRANGEMENT & ELEVATION	PROCESS EQUIPMENT DESIGN & DRAWING	B.E. CHEMICAL ENGINEERING	1:120 METRIC
STUDENT NAME	**ROLL NO.**	**DRAWING NO.**	**DATE**
SOURISH SENAPATI	002310301035	JU/ChE/ED/DWG-03	01 / 09 / 2026

ETHANOL DISTILLATION (C-201) | 700 kmol/h FEED (7 mol% EtOH) | ID 1.20 m, R=3.0 | ASME SEC VIII / TEMA